

Non-Pseudo-Arithmetic Hyperbolic 4-Manifolds

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Abstract

We give three explicit finite-volume hyperbolic Coxeter lattices in dimension four whose adjoint trace fields are not totally real. They are the reflection groups of the Ma–Zheng polytopes $P_{11,6}, P_{11,7}, P_{11,8}$. The obstruction is contained in two cyclic products of the doubled unit-normal Gram matrix $A = 2G$. Those products recover elements $s = \sqrt{5}$ and u satisfying

$$(s - 1)u^2 + 2(s - 3)u + s - 7 = 0.$$

After the embedding $s \mapsto -\sqrt{5}$, this quadratic has discriminant $8(1 - \sqrt{5}) < 0$. The cyclic-product field therefore has a complex place. Vinberg’s identification of that field with the adjoint trace field and the Emery–Mila total-reality obstruction imply that the three lattices are not pseudo-arithmetic. Torsion-free orientation-preserving finite-index subgroups yield orientable noncompact finite-volume hyperbolic 4-manifolds with the same obstruction.

The exposition includes all specialized definitions used in the argument, an exact source-to-matrix transcription, a proof of every new algebraic step, independent exact certificates, and an explicit register of imported theorems. A pinned Lean 4 development kernel-checks the radical identity, the number-field obstruction, the main implication, and the finite-index corollary. Statements imported from geometry and the literature are represented in Lean as local hypotheses rather than asserted as axioms or reproved here.

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1 Introduction and complete reader's primer

1.1 The question and the answer

A *lattice* in the isometry group of real hyperbolic n -space is a discrete group whose quotient has finite hyperbolic volume. Arithmetic lattices are constructed from algebraic groups over number fields. Many important nonarithmetic lattices are assembled by gluing arithmetic pieces. Emery and Mila introduced *pseudo-arithmeticity* to give one common algebraic framework for arithmetic lattices, quasi-arithmetic lattices, and the familiar gluing constructions [2]. They asked whether every lattice in $\mathrm{PO}(n, 1)$, for $n > 3$, is pseudo-arithmetic and, more weakly, whether every such lattice has totally real adjoint trace field [2, Questions 1.8 and 1.10].

Both questions have negative answers in dimension four.

Theorem 1.1 (Main theorem). *For $j \in \{6, 7, 8\}$, let $P_{11,j}$ be the finite-volume hyperbolic Coxeter 4-polytope in the Ma–Zheng census recorded in Table 1, and let $\Gamma_{11,j} < \mathrm{PO}(4, 1)$ be its reflection group. Then the adjoint trace field of $\Gamma_{11,j}$ is not totally real. Consequently $\Gamma_{11,j}$ is not pseudo-arithmetic.*

Corollary 1.2 (Manifold form). *There exist orientable, noncompact, finite-volume hyperbolic 4-manifolds whose fundamental groups are not pseudo-arithmetic.*

Ma and Zheng classify all 331 finite-volume hyperbolic Coxeter 4-polytopes with seven facets [5]. The present examples are three consecutive entries in that census. The proof needs only four entries of their Gram matrices and one common radical. It does not require a numerical approximation, a primitive element for the entire field, or a classification of all embeddings of that field.

Emery–Mila checked the higher-dimensional Coxeter examples then available to them. The final 2026 version of Belolipetsky–Bogachev–Kolpakov–Slavich still described the known examples in $\text{PO}(n, 1)$, $n > 3$, as pseudo-arithmetic [1]. A targeted literature audit through 24 September 2026 found no earlier recorded non-pseudo-arithmetic lattice in these dimensions. We therefore use the qualified statement that we know of no previous example; a bounded search is not a proof of absolute priority.

1.2 All specialized language used in the proof

This subsection is a compact primer. Later sections repeat the definitions at the point where calculations use them.

Hyperbolic space and lattices. On $\mathbb{R}^{n+1}$, put

$$\langle x, y \rangle_{n,1} = x_1 y_1 + \cdots + x_n y_n - x_{n+1} y_{n+1}.$$

The upper sheet

$$\mathbb{H}^n = \{x \in \mathbb{R}^{n+1} : \langle x, x \rangle_{n,1} = -1, x_{n+1} > 0\}$$

is the hyperboloid model of real hyperbolic space. The group preserving the form acts by isometries; after quotienting by its scalar center one obtains the projective orthogonal group $\text{PO}(n, 1)$. A subgroup $\Gamma < \text{Isom}(\mathbb{H}^n)$ is *discrete* if the identity is isolated in Γ , and it is a *lattice* if $\Gamma \backslash \mathbb{H}^n$ has finite volume. The quotient is an orbifold in general and a manifold when Γ is torsion-free.

Coxeter polytopes and Gram matrices. A convex hyperbolic polytope is *Coxeter* when every dihedral angle is π/m for an integer $m \geq 2$. Reflections in its facets generate a discrete group for which the polytope is a fundamental domain. If the polytope has finite volume, this reflection group is a lattice. Choose an outward unit spacelike normal e_i to each facet. The matrix $G = (\langle e_i, e_j \rangle_{n,1})$ is the *unit-normal Gram matrix*. Its diagonal entries are one. Off the diagonal one has

$$g_{ij} = \begin{cases} -\cos(\pi/m_{ij}), & \text{the facets meet at angle } \pi/m_{ij}, \\ -1, & \text{the facets are parallel,} \\ -\cosh(\rho_{ij}), & \text{they are disjoint at distance } \rho_{ij}. \end{cases}$$

We use $A = 2G$. A *cyclic product* is a product $A_{i_0 i_1} A_{i_1 i_2} \cdots A_{i_m i_0}$. The field generated over $\mathbb{Q}$ by every such product is the *Vinberg field* $k(P)$.

Number fields and total reality. A *number field* is a finite extension of $\mathbb{Q}$. An embedding of a number field K is an injective field homomorphism $K \hookrightarrow \mathbb{C}$ fixing $\mathbb{Q}$. The field K is *totally real* when every such embedding has image in $\mathbb{R}$. Thus one nonreal embedding is a complete certificate that K is not totally real. We use the standard extension theorem: if $L \subset K$ are number fields, every embedding $L \hookrightarrow \mathbb{C}$ extends to an embedding $K \hookrightarrow \mathbb{C}$. This follows from the existence of a complex root of the transported minimal polynomial; a proof is recalled in Proposition 2.3.

Adjoint trace fields. For a matrix Lie group, conjugation differentiates to a linear action Ad on its Lie algebra. The *adjoint trace field* of a lattice Γ is

$$k(\Gamma) = \mathbb{Q}(\{\text{tr}(\text{Ad } \gamma) : \gamma \in \Gamma\}) \subset \mathbb{R}.$$

It is unchanged on passing to a finite-index subgroup. Vinberg’s field-of-definition theorem identifies this field, for a finite-volume Coxeter reflection group, with the cyclic-product field $k(P)$. This identification is an imported theorem, not a new result proved here.

Pseudo-arithmeticity. Vinberg associates to a Zariski-dense lattice Γ an algebraic group over $k(\Gamma)$, called its ambient group. An algebraic group over a totally real field k is *admissible* for hyperbolic n -space when one real factor is the required noncompact projective orthogonal group and the other real factors are compact. In Emery–Mila’s definition, a lattice is *pseudo-arithmetic* when its ambient group is obtained from such an admissible group by scalar extension through a finite real multiquadratic extension K/k . Their definition implies that $K = k(\Gamma)$ is totally real. This last necessary condition is the only part of pseudo-arithmeticity used in the proof.

Formal verification terminology. Lean checks a theorem only from its declarations, imported library theorems, and explicitly supplied hypotheses. In this release, the new algebraic argument is proved inside Lean. A literature result such as Vinberg’s theorem is represented by a named hypothesis with the same logical type needed by the manuscript. Lean checks every implication after that boundary, but this does not constitute a new formal proof of Vinberg’s theorem. No project-defined `axiom`, `opaque`, `sorry`, or `admit` is used.

1.3 The certificate in one paragraph

Let G be any of the three unit-normal Gram matrices and put $A = 2G$. With zero-based facet indices, the common entries

$$A_{12} = -\frac{1 + \sqrt{5}}{2}, \quad A_{03} = -1, \quad A_{35} = -\sqrt{2}, \quad A_{50} = -2a$$

give two cyclic products

$$c_{12} = A_{12}A_{21} = \frac{3 + \sqrt{5}}{2}, \quad c_{035} = A_{03}A_{35}A_{50} = -2\sqrt{2}a.$$

Hence the Vinberg field contains $s = 2c_{12} - 3 = \sqrt{5}$ and $u = -c_{035}/2 = \sqrt{2}a$. The exact radical for a implies

$$(s - 1)u^2 + 2(s - 3)u + s - 7 = 0.$$

At the conjugate embedding of $\mathbb{Q}(\sqrt{5})$, this quadratic has negative discriminant. It therefore produces a nonreal embedding of $\mathbb{Q}(s, u)$, which extends to the whole Vinberg field. This is the entire new obstruction.

1.4 Logical status of the ingredients

For clarity, the proof has four layers.

- L1.** *Elementary proof:* the matrix entries, two products, radical elimination, discriminant, and complex-place deduction are written out in Sections 4–6.

- L2.** *Exact computation:* independent Python and PARI/GP programs check the source transcription, rank, inertia, field signatures, and finite-volume combinatorics. They support the input data but are not needed to trust the two-cycle algebra once the displayed data are accepted.
- L3.** *Lean kernel checking:* the internal implication from the radical formula and cycle certificate to non-total reality and non-pseudo- arithmeticity is formalized. The finite-index corollary is checked from explicitly named geometric hypotheses.
- L4.** *External mathematics:* the census classification, Vinberg’s field identification, the Emery–Mila necessary condition, Selberg’s lemma, and finite-index invariance are cited rather than reproved. The exact list appears in Section 9.

This separation prevents an executable check of a matrix identity from being misdescribed as a formal verification of a classification theorem.

2 Basic field and hyperbolic facts

2.1 Embeddings, minimal polynomials, and complex places

We record the elementary field facts at the level needed later.

Definition 2.1. Let K be a number field and $\alpha \in K$. The *minimal polynomial* $\text{minpoly}_{\mathbb{Q}}(\alpha)$ is the unique monic polynomial of least positive degree in $\mathbb{Q}[X]$ that vanishes at α .

The uniqueness follows from Euclidean division: if two monic polynomials of least degree vanish, their difference has smaller degree and also vanishes. The kernel of the evaluation homomorphism $\mathbb{Q}[X] \rightarrow K, p \mapsto p(\alpha)$, is therefore generated by this polynomial.

Lemma 2.2. *The number $\sqrt{5}$ is irrational and*

$$\text{minpoly}_{\mathbb{Q}}(\sqrt{5}) = X^2 - 5.$$

Its two embeddings over $\mathbb{Q}$ send $\sqrt{5}$ to $\sqrt{5}$ and $-\sqrt{5}$, respectively.

Proof. Suppose $\sqrt{5} = p/q$ for coprime positive integers p, q . Then $p^2 = 5q^2$, so the prime 5 divides p^2 , hence divides p . Write $p = 5r$. Substitution gives $q^2 = 5r^2$, so $5 \mid q$, contrary to coprimality. Thus $\sqrt{5} \notin \mathbb{Q}$. It is a root of the monic quadratic $X^2 - 5$; irrationality rules out a linear minimal polynomial, so this quadratic is minimal. Any embedding preserves addition, multiplication, and rational numbers, and hence must send $\sqrt{5}$ to a root of $X^2 - 5$. The two roots are $\pm\sqrt{5}$, and both choices define an embedding of $\mathbb{Q}(\sqrt{5})$. $\square$

Proposition 2.3 (Extension of embeddings). *Let $L \subset K$ be number fields and let $\sigma : L \hookrightarrow \mathbb{C}$ be an embedding. Then σ extends to an embedding $\tilde{\sigma} : K \hookrightarrow \mathbb{C}$.*

Proof. This is the standard embedding-extension theorem; we recall its mechanism. Because number fields have characteristic zero, K/L is finite and separable. By the primitive-element theorem there is $\theta \in K$ with $K = L(\theta)$. Let $p \in L[X]$ be the minimal polynomial of θ . Apply σ coefficientwise to obtain $p^\sigma \in \mathbb{C}[X]$. The fundamental theorem of algebra supplies a root $z \in \mathbb{C}$ of p^σ . The map

$$L[X] \longrightarrow \mathbb{C}, \quad \sum_i a_i X^i \longmapsto \sum_i \sigma(a_i) z^i$$

has (p) in its kernel and therefore descends to a homomorphism $L[X]/(p) \rightarrow \mathbb{C}$. Identifying $L[X]/(p)$ with $L(\theta) = K$ gives the extension. A unital homomorphism from a field has zero kernel, so it is an embedding. $\square$

Lemma 2.4 (Quadratic real-root test). *Let $a, b, c \in \mathbb{R}$, with $a \neq 0$, and put $D = b^2 - 4ac$. The equation $ax^2 + bx + c = 0$ has a real solution if and only if $D \geq 0$.*

Proof. Multiplying the equation by $4a$ and completing the square gives

$$(2ax + b)^2 = b^2 - 4ac = D.$$

A real square is nonnegative, proving necessity. If $D \geq 0$, the two values $(-b \pm \sqrt{D})/(2a)$ are real and substitution shows that they are solutions. $\square$

Proposition 2.5 (A nonreal subfield embedding persists). *Let $L \subset K$ be number fields. If L has a nonreal embedding, then K is not totally real.*

Proof. Let $\sigma : L \hookrightarrow \mathbb{C}$ have image not contained in $\mathbb{R}$. Proposition 2.3 extends it to $\tilde{\sigma} : K \hookrightarrow \mathbb{C}$. If the image of $\tilde{\sigma}$ were contained in $\mathbb{R}$, then so would be the image of its restriction σ , a contradiction. $\square$

2.2 The hyperboloid model and reflecting hyperplanes

Let $V = \mathbb{R}^{n+1}$ with the form $\langle \cdot, \cdot \rangle_{n,1}$ defined in Subsection 1.2. A vector is spacelike, null, or timelike according as its norm is positive, zero, or negative. If e is a unit spacelike vector, then

$$H_e = \{x \in \mathbb{H}^n : \langle x, e \rangle_{n,1} = 0\}$$

is a totally geodesic hyperplane, and

$$r_e(x) = x - 2\langle x, e \rangle_{n,1}e$$

is the reflection fixing H_e pointwise. Indeed, $r_e(e) = -e$, r_e fixes the orthogonal complement of e , and direct expansion shows $\langle r_e(x), r_e(y) \rangle_{n,1} = \langle x, y \rangle_{n,1}$.

For a polytope described as the intersection of half-spaces $\langle x, e_i \rangle_{n,1} \leq 0$, the vectors e_i are outward normals. The inner products of pairs of normals give the Gram formulas in Subsection 1.2; these are the standard two-dimensional Lorentzian calculations for intersecting, asymptotic, and ultraparallel hyperplanes [6, Chapter 3].

Definition 2.6. A Coxeter matrix has diagonal entries one and off-diagonal labels $m_{ij} \in \{2, 3, \dots\} \cup \{\infty\}$. Its Coxeter group is generated by involutions r_i with relations $(r_i r_j)^{m_{ij}} = 1$ when $m_{ij} < \infty$. For a hyperbolic Coxeter polytope, these generators are the geometric facet reflections.

The fact that the geometric reflection group is discrete and that the polytope is a fundamental domain is the Poincaré–Coxeter theorem. We use it as standard background. Finite volume of the fundamental domain is exactly finite covolume of the group.

2.3 From an orbifold to a manifold

Two elementary finite-index observations will be used. First, the kernel of the orientation character $\Gamma \rightarrow \{\pm 1\}$ has index at most two. For a group generated by hyperplane reflections, each generator reverses orientation, so the kernel has index exactly two. Second, a subgroup of finite index d gives a degree- d orbifold cover and multiplies volume by d . Thus finite volume is preserved in both directions.

The following external theorem removes torsion.

Theorem 2.7 (Selberg’s lemma). *Every finitely generated subgroup of $\mathrm{GL}_N(F)$, where F has characteristic zero, has a torsion-free subgroup of finite index.*

We cite Selberg’s original result [7]. A Coxeter group with finitely many facets is finitely generated, and its reflection representation is linear over a characteristic-zero field, so the theorem applies. A torsion-free discrete group acts freely on $\mathbb{H}^n$: an element fixing a point lies in a compact point stabilizer, and discreteness makes the cyclic subgroup finite, hence trivial. The quotient is therefore a manifold. Passing inside the orientation-preserving subgroup makes it orientable. Finally, a finite cover of a noncompact finite-volume quotient is noncompact: if the cover were compact, its continuous image would be compact, contrary to noncompactness of the base.

3 Adjoint trace fields and pseudo-arithmeticity

3.1 The adjoint representation and ambient group

Let $\mathbf{G}$ be a real algebraic matrix group with Lie algebra $\mathfrak{g}$. Conjugation by $g \in \mathbf{G}(\mathbb{R})$ maps $X \in \mathfrak{g}$ to gXg^{-1} . This linear map is $\mathrm{Ad}(g)$. Its trace is independent of a choice of basis for $\mathfrak{g}$. For a Zariski-dense subgroup Γ , define

$$k(\Gamma) = \mathbb{Q}(\mathrm{tr}(\mathrm{Ad} \gamma) : \gamma \in \Gamma).$$

For lattices in a connected noncompact simple Lie group, Borel density gives the required Zariski density. Vinberg’s theory produces a form of the ambient algebraic group over $k(\Gamma)$, unique in the appropriate isomorphism or isogeny class [8]. We do not need its construction; only the field is used below.

We also use the standard finite-index invariance

$$[\Gamma : \Lambda] < \infty \implies k(\Lambda) = k(\Gamma). \quad (3.1)$$

This is part of Vinberg’s commensurability theory. One inclusion is immediate from $\Lambda \subset \Gamma$; equality is the nontrivial external statement and is listed as Input E6 in Section 9.

3.2 Pseudo-admissibility

An algebraic k -group $\mathbf{H}$ is *admissible* for $\mathrm{PO}(n, 1)$ when k is totally real and, among the groups obtained from the real embeddings of k , exactly one has the required noncompact factor $\mathrm{PO}(n, 1)^\circ$, while the others are compact. This is the usual arithmetic admissibility condition.

Definition 3.1 (Emery–Mila). Let $n > 3$. A group over a number field K is *pseudo-admissible* over K/k if it is the scalar extension $\mathbf{H}_K$ of an admissible k -group through a finite, possibly trivial, real multiquadratic extension

$$K = k(\sqrt{d_1}, \dots, \sqrt{d_t}).$$

A lattice $\Gamma < \mathrm{PO}(n, 1)$ is *pseudo-arithmetic* when its ambient group is pseudo-admissible [2, Definitions 1.6–1.7].

The trivial case $t = 0$ includes quasi-arithmetic lattices. The adjective “real” in this definition is the archimedean condition ensuring that the extension has only real embeddings over the relevant real places.

Proposition 3.2 (Total-reality obstruction). *If $n > 3$ and $\Gamma < \mathrm{PO}(n, 1)$ is pseudo-arithmetic, then its adjoint trace field $k(\Gamma)$ is totally real.*

Proof. This is the immediate necessary condition recorded by Emery–Mila [2, Section 1.4]. In their definition the base field is totally real by admissibility and K/k is a real multiquadratic extension. Thus every embedding $K \hookrightarrow \mathbb{C}$ has image in $\mathbb{R}$. Since $K = k(\Gamma)$, the adjoint trace field is totally real. $\square$

The contrapositive is what we use:

$$k(\Gamma) \text{ not totally real} \implies \Gamma \text{ not pseudo-arithmetic.} \quad (3.2)$$

3.3 Vinberg’s cyclic-product field

Let $P \subset \mathbb{H}^n$ be a finite-volume Coxeter polytope with unit-normal Gram matrix $G = (g_{ij})$, and put $A = 2G$. For a closed sequence of facet indices $i_0, i_1, \dots, i_m, i_0$, its cyclic product is

$$A_{i_0 i_1} A_{i_1 i_2} \cdots A_{i_{m-1} i_m} A_{i_m i_0}.$$

Repeated indices are allowed, and reversing the cycle gives the same value because A is symmetric. Define

$$k(P) = \mathbb{Q}(\text{all cyclic products in } A).$$

Theorem 3.3 (Vinberg). *If P has finite volume and $\Gamma(P)$ is its reflection group, then*

$$k(\Gamma(P)) = k(P).$$

This is the cyclic-product form of Vinberg’s field-of-definition theorem [8]; see also [2, Section 4.1] and [4, Section 3.1]. It is an external input. The factor two is not cosmetic: Vinberg’s convention uses the bilinear Gram matrix with diagonal two. Starting from unit normals, that matrix is $A = 2G$.

Remark 3.4 (Entry field versus trace field). The field generated by all individual entries of G can be strictly larger than $k(P)$. An individual entry changes if a normal vector is rescaled, whereas closed products cancel these rescalings. Accordingly, showing that a radical occurs in one Gram entry does not show that it lies in the adjoint trace field. Every field inclusion below is obtained from an explicit closed product.

4 The three Coxeter polytopes and their exact matrices

4.1 How the census vector is read

We number the seven facets $0, 1, \dots, 6$. The 21 unordered pairs are listed in lexicographic blocks:

$$\begin{aligned} (01, 02, 03, 04, 05, 06) \mid (12, 13, 14, 15, 16) \mid (23, 24, 25, 26) \\ \mid (34, 35, 36) \mid (45, 46) \mid (56). \end{aligned} \quad (4.1)$$

An integer $m \geq 2$ means dihedral angle π/m , the symbol 0 means parallel facets, and a letter means an ultraparallel pair. Thus the initial block $2232ab$ means

$$m_{01} = 2, \quad m_{02} = 2, \quad m_{03} = 3, \quad m_{04} = 2, \quad g_{05} = -a, \quad g_{06} = -b.$$

The relevant published rows are as follows. The displayed volume is the volume of the Coxeter fundamental polytope.

The assertion that these rows define finite-volume polytopes is external Input E1. Section 7 describes independent exact checks, including a direct finite-volume check for $P_{11,8}$.

polytope	Coxeter vector	cusps	volume
$P_{11,6}$	2232ab 52322 2220 442 22 2	1	$13\pi^2/864$
$P_{11,7}$	2232ab 52323 2220 442 22 2	1	$\pi^2/54$
$P_{11,8}$	2232ab 52324 2220 442 22 2	2	$19\pi^2/864$

Table 1: The three Ma–Zheng census rows used in the proof.

4.2 Radicals and a uniform matrix

Put

$$s = \sqrt{5}, \quad r = \sqrt{2},$$

and let $q_6 = 2, q_7 = 3, q_8 = 4$. The common first dotted entry is

$$a = \frac{1}{2}\sqrt{7 + s + 2\sqrt{2 + 2s}}. \quad (4.2)$$

The second dotted entries are

$$\begin{aligned} b_6 &= \sqrt{\frac{9}{2} + 2s + \sqrt{38 + 17s}}, \\ b_7 &= r + \frac{\sqrt{10}}{2} + \frac{1}{2}\sqrt{29 + 13s}, \\ b_8 &= \frac{1}{4}(2 + 3r + 2s + rs + 2\sqrt{19 + 14r + 9s + 6rs}). \end{aligned} \quad (4.3)$$

All square roots mean the positive real root. Their radicands are positive, and the resulting a, b_j exceed one, as required for ultraparallel facets. Only a is used in the obstruction.

For reference, the complete unit-normal Gram matrix is

$$G_j = \begin{pmatrix} 1 & 0 & 0 & -\frac{1}{2} & 0 & -a & -b_j \\ 0 & 1 & -\frac{1+s}{4} & 0 & -\frac{1}{2} & 0 & -\cos(\pi/q_j) \\ 0 & -\frac{1+s}{4} & 1 & 0 & 0 & 0 & -1 \\ -\frac{1}{2} & 0 & 0 & 1 & -\frac{r}{2} & -\frac{r}{2} & 0 \\ 0 & -\frac{1}{2} & 0 & -\frac{r}{2} & 1 & 0 & 0 \\ -a & 0 & 0 & -\frac{r}{2} & 0 & 1 & 0 \\ -b_j & -\cos(\pi/q_j) & -1 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (4.4)$$

To check the transcription, read each row against (4.1). For example, $m_{12} = 5$ gives $g_{12} = -\cos(\pi/5) = -(1 + s)/4$; $m_{26} = 0$ gives $g_{26} = -1$; and $m_{34} = m_{35} = 4$ gives $-r/2$. The elementary special-angle identities used here are

$$\cos(\pi/3) = \frac{1}{2}, \quad \cos(\pi/4) = \frac{\sqrt{2}}{2}, \quad \cos(\pi/5) = \frac{1 + \sqrt{5}}{4}.$$

The first two follow from an equilateral triangle and an isosceles right triangle. For the third, put $z = e^{i\pi/5}$ and $x = z + z^{-1} = 2\cos(\pi/5)$. Since $z^5 = -1$ and $z \neq -1$, division of $z^5 + 1$ by $z + 1$ gives $z^4 - z^3 + z^2 - z + 1 = 0$. After division by z^2 , this becomes

$$(z^2 + z^{-2}) - (z + z^{-1}) + 1 = 0.$$

Because $z^2 + z^{-2} = x^2 - 2$, we get $x^2 - x - 1 = 0$. Positivity selects $x = (1 + \sqrt{5})/2$.

4.3 The two cyclic products

Put $A_j = 2G_j$. Four common entries of A_j are

$$A_{12} = -\frac{1+s}{2}, \quad A_{03} = -1, \quad A_{35} = -r, \quad A_{50} = -2a. \quad (4.5)$$

The first closed path has two directed edges, $1 \rightarrow 2 \rightarrow 1$, and gives

$$\begin{aligned} c_{12} &= A_{12}A_{21} = \left(-\frac{1+s}{2}\right)^2 \\ &= \frac{1+2s+s^2}{4} = \frac{6+2s}{4} = \frac{3+s}{2}, \end{aligned} \quad (4.6)$$

where $s^2 = 5$. Thus

$$s = 2c_{12} - 3 \in k(P_{11,j}). \quad (4.7)$$

The second closed path is $0 \rightarrow 3 \rightarrow 5 \rightarrow 0$:

$$c_{035} = A_{03}A_{35}A_{50} = (-1)(-r)(-2a) = -2ra. \quad (4.8)$$

Consequently

$$u := ra = -\frac{1}{2}c_{035} \in k(P_{11,j}). \quad (4.9)$$

These equations prove membership in the cyclic-product field directly. No individual-entry argument is being used.

5 The radical identity and the complex place

5.1 Deriving the quadratic relation

We now prove the one algebraic identity behind all three cases.

Lemma 5.1 (Exact radical elimination). *Let $s = \sqrt{5}$, let a be given by (4.2), and let $u = \sqrt{2}a$. Then*

$$(s-1)u^2 + 2(s-3)u + s - 7 = 0. \quad (5.1)$$

Proof. Set

$$t = \sqrt{2+2s} > 0.$$

By the definitions of a and u ,

$$u^2 = 2a^2 = 2 \cdot \frac{1}{4}(7+s+2t) = \frac{7+s}{2} + t. \quad (5.2)$$

We claim first that

$$(s-1)u = 3 - s + t. \quad (5.3)$$

Both sides are positive: $s > 1$ and $u > 0$ for the left side, while $s < 3$ and $t > 0$ for the right side. It is therefore enough to prove that their squares are equal. Using $s^2 = 5$, $t^2 = 2 + 2s$, and (5.2), expansion gives

$$\begin{aligned} ((s-1)u)^2 &= (s-1)^2 \left(\frac{7+s}{2} + t \right) \\ &= (6-2s) \left(\frac{7+s}{2} + t \right) \\ &= 16 - 4s + (6-2s)t, \end{aligned}$$

because $(6 - 2s)(7 + s)/2 = (42 - 8s - 2s^2)/2 = 16 - 4s$. On the other hand,

$$\begin{aligned}(3 - s + t)^2 &= (3 - s)^2 + 2(3 - s)t + t^2 \\ &= 14 - 6s + 2(3 - s)t + 2 + 2s \\ &= 16 - 4s + (6 - 2s)t.\end{aligned}$$

The positive quantities have equal squares, proving (5.3).

Rearrange it as

$$t = (s - 1)u + s - 3. \tag{5.4}$$

Squaring, substituting $t^2 = 2 + 2s$, and moving everything to the left gives

$$((s - 1)u + s - 3)^2 - (2 + 2s) = 0.$$

A direct expansion factors the left side as

$$(s - 1)((s - 1)u^2 + 2(s - 3)u + s - 7).$$

Because $s \neq 1$, division by $s - 1$ proves the claimed relation. $\square$

Remark 5.2. The expanded proof uses only positivity of the named square roots, $s^2 = 5$, and ring arithmetic. The Lean theorem

`explicit_radical_cycle_relation`

checks the same identity from the nested square-root definitions. The intermediate theorem

`cycle_relation_of_radical_identities`

checks the final factorization over an arbitrary characteristic-zero field.

5.2 Negative discriminant at the conjugate place

Lemma 5.3 (Non-total-reality certificate). *Let $s = \sqrt{5}$ and let u satisfy (5.1). Then $L = \mathbb{Q}(s, u)$ is not totally real.*

Proof. View (5.1) as $f(u) = 0$, where

$$f(X) = (s - 1)X^2 + 2(s - 3)X + s - 7 \in \mathbb{Q}(s)[X].$$

Its leading coefficient is nonzero because $s \neq 1$. Its discriminant is

$$\begin{aligned}\Delta &= (2(s - 3))^2 - 4(s - 1)(s - 7) \\ &= 4((s - 3)^2 - (s - 1)(s - 7)) \\ &= 4((s^2 - 6s + 9) - (s^2 - 8s + 7)) \\ &= 8(1 + s).\end{aligned} \tag{5.5}$$

Let $\sigma : \mathbb{Q}(s) \hookrightarrow \mathbb{R}$ be the nonidentity embedding, so $\sigma(s) = -s$. Applying σ to the coefficients gives a real quadratic f^σ with discriminant

$$\sigma(\Delta) = 8(1 - s) < 0, \tag{5.6}$$

because $s = \sqrt{5} > 1$. By Lemma 2.4, f^σ has no real root.

If $u \in \mathbb{Q}(s)$, then applying σ to $f(u) = 0$ would make the real number $\sigma(u)$ a root of f^σ , a contradiction. Hence $u \notin \mathbb{Q}(s)$. Since f is quadratic and has u as a root, it is the minimal polynomial of u over $\mathbb{Q}(s)$. Extend σ from $\mathbb{Q}(s)$ to $L = \mathbb{Q}(s, u)$. The image of u must be one of the two roots of f^σ ; neither root is real. Thus this extension is a nonreal embedding of L , and L is not totally real. $\square$

There is a shorter intrinsic formulation. If a number field K contains s, u with

$$\text{minpoly}_{\mathbb{Q}}(s) = X^2 - 5, \quad (s - 1)u^2 + 2(s - 3)u + s - 7 = 0,$$

then the root $-\sqrt{5}$ of the minimal polynomial is realized as the image of s under some embedding $K \hookrightarrow \mathbb{C}$. If that embedding were real, the image of u would solve the impossible conjugate quadratic. This is exactly the formulation kernel-checked in `not_totallyReal_of_cycle_minpoly`.

6 Proofs of the main theorem and manifold corollary

Proof of Theorem 1.1. Fix $j \in \{6, 7, 8\}$. Let $\Gamma_j = \Gamma(P_{11,j})$ be the group generated by reflections in the facets of $P_{11,j}$. By the Ma–Zheng census statement, $P_{11,j}$ has finite volume. Therefore Γ_j is a lattice.

Let $K_j = k(\Gamma_j)$ be its adjoint trace field. By Theorem 3.3,

$$K_j = k(P_{11,j}). \tag{6.1}$$

The products in (4.6) and (4.8) are cyclic products of $A_j = 2G_j$. Equations (4.7) and (4.9) therefore show that

$$s = \sqrt{5} \in K_j, \quad u = \sqrt{2}a \in K_j.$$

Consequently $L = \mathbb{Q}(s, u)$ is a subfield of K_j .

Lemma 5.1 gives the quadratic relation for s, u , and Lemma 5.3 gives a nonreal embedding of L . By Proposition 2.5, that embedding extends to a nonreal embedding of K_j . Thus K_j is not totally real.

If Γ_j were pseudo-arithmetic, Proposition 3.2 would make K_j totally real. This contradicts the preceding paragraph. Hence Γ_j is not pseudo-arithmetic. The computation was independent of j , so it applies to all three rows. $\square$

Proof of Corollary 1.2. Choose one j and write $\Gamma = \Gamma_{11,j}$. Let $\Gamma^+ = \Gamma \cap \text{Isom}^+(\mathbb{H}^4)$. The orientation character shows that $[\Gamma : \Gamma^+] = 2$. The group Γ^+ is finitely generated and linear in characteristic zero. By Selberg’s lemma, it contains a torsion-free subgroup Λ of finite index.

The action of Λ on $\mathbb{H}^4$ is free and orientation preserving, so

$$M = \Lambda \backslash \mathbb{H}^4$$

is an orientable hyperbolic 4-manifold. Since Λ has finite index in the lattice Γ , its quotient has finite volume. The base orbifold is noncompact because the census row has at least one ideal vertex; the finite cover M is therefore noncompact as explained in Subsection 2.3.

Finite-index invariance of the adjoint trace field gives

$$k(\Lambda) = k(\Gamma) = K_j.$$

Theorem 1.1 says that this field is not totally real. Applying Proposition 3.2 to Λ shows that Λ , and equivalently the manifold with this fundamental group, is not pseudo-arithmetic. $\square$

7 Exact computational certification

The preceding proof is hand-checkable after accepting the source rows and the two external field theorems. The repository also contains redundant exact checks. Their purpose is to detect transcription mistakes and to make every concrete assertion reproducible.

7.1 Rank and inertia of all three Gram matrices

Exact symbolic elimination gives

$$\text{rank}(G_j) = 5, \quad \text{inertia}(G_j) = (4, 1, 2)$$

for $j = 6, 7, 8$. Here the entries count positive, negative, and zero eigenvalues. The common principal block on facets $0, 1, 2, 3, 4$ has leading principal minors

$$1, \text{quad}, \text{quad} \frac{5-s}{8}, \text{quad} \frac{3(5-s)}{32}, \text{quad} -\frac{1+s}{32}. \quad (7.1)$$

They are all nonzero because $1 < s < 3$. The signs are $+, +, +, +, -$; the Jacobi signature criterion therefore gives four positive and one negative direction on this block. Exact elimination shows that the full matrix has rank five, so the remaining two directions are null. The programs additionally verify that the determinant and every principal six-by-six minor vanish. The nonzero five-by-five minor and the elimination certificate are the rank witnesses; no floating-point eigenvalue is used.

7.2 Direct finite-volume check for the third polytope

For $P_{11,8}$, the script enumerates all 127 nonempty principal submatrices and matches them with the ten maximal vertex subdiagrams of the published combinatorial type. Eight vertices have elliptic rank-four links. The remaining two are parabolic:

$$\{1, 3, 4, 6\} \text{ of type } \tilde{C}_3, \quad \{2, 3, 4, 5, 6\} \text{ of type } \tilde{A}_1 \sqcup \tilde{C}_2.$$

The second has nullity two and is the Euclidean triangular-prism link of a nonsimple ideal vertex. Every elliptic rank-three edge diagram has exactly two ordinary or ideal endpoints. Vinberg's finite-volume criterion [9] then gives finite volume and exactly two cusps.

An independent CoxIter run, pinned to a recorded source commit, returns

$$(f_0, f_1, f_2, f_3, f_4) = (10, 21, 18, 7, 1), \quad \text{vol}(P_{11,8}) = \frac{19\pi^2}{864},$$

agreeing with the census [3]. This redundancy verifies the geometric input for one case; the proof of the family still cites the Ma–Zheng classification for all three.

7.3 A complete trace-field check for the third polytope

The two cycles used in the theorem give only the subfield needed for the obstruction. Two further cycles recover $r = \sqrt{2}$ and b_8 :

$$c_{126} = -r(1+s), \quad c_{03416} = -4b_8.$$

Since every Gram entry lies in $\mathbb{Q}(s, r, a, b_8)$, this proves

$$k(P_{11,8}) = \mathbb{Q}(s, r, a, b_8).$$

With $w = 2(r + a + b_8)$, exact power-basis reconstruction gives $k(P_{11,8}) = \mathbb{Q}(w)$, where

$$p_w(X) = X^8 - 8X^7 - 252X^6 - 1392X^5 - 3524X^4 - 5280X^3 - 5712X^2 - 3968X - 1136. \quad (7.2)$$

PARI/GP verifies irreducibility, signature $(r_1, r_2) = (4, 2)$, and discriminant

$$40960000 = 2^{16}5^4.$$

Thus the full field independently exhibits two conjugate pairs of complex embeddings. A second primitive-element route produces a different polynomial for the same explicitly reconstructed field; agreement is not inferred from degree, signature, and discriminant alone.

7.4 Artifacts and bounded-memory execution

The exact programs and durable outputs are:

- `scripts/p11_6_7_non_total_trace.py`;
 `results/p11_6_7_non_total_trace_certificate.json`;
- `scripts/p11_8_non_total_trace.py`;
 `results/p11_8_non_total_trace_certificate.json`;
- `scripts/p118_trace_field.py`;
 `results/p118_trace_field_certificate.json`;
- `scripts/p118_field_independent.gp`, the independent PARI/GP irreducibility, real-root, signature, and discriminant check;
- `scripts/p11_8_finite_volume.py`;
 `results/p11_8_finite_volume_validation.json`;
- the pinned CoxIter input, output, and provenance files under `results/`.

Each expensive symbolic case runs in a fresh process. On the reference machine the case-worker resident-set peaks are below 100 MB. The exact commands, source URLs, versions, and expected digests are in `REPRODUCIBILITY.md`.

The Ma–Zheng source file used by the certifiers has SHA-256

$$0017931637f017641d4935450cc2a07d070866d9e3ede9f3a39f003.$$

Each headline Python certifier checks this digest before parsing the rows.

8 Lean formalization and trust boundary

The directory `formal/` is a Lean 4 project pinned to Lean and `mathlib v4.32.0`. This section states exactly what its successful build means.

8.1 Unconditional kernel-checked mathematics

The module `PseudoArithmetic/Manuscript.lean` proves the following.

1. `recover_s_from_twoCycle` and `recover_u_from_threeCycle` check the two rearrangements in (4.7) and (4.9) over every characteristic-zero field.
2. `explicit_radical_cycle_relation` defines the positive nested radicals s, t, a, u over $\mathbb{R}$, proves their square and positivity facts, and derives (5.1).
3. `conjugateQuadratic_no_real_root` proves directly that the conjugate relation has no real solution when $s^2 = 5$ and $s < 0$.
4. `not_totallyReal_of_cycle_minpoly` uses mathlib’s number-field embedding theorem to obtain an embedding sending s to $-\sqrt{5}$, proves that embedding nonreal, and concludes that the field is not totally real.
5. A `TwoCycleCertificate` packages exactly the four algebraic values and two equations used by the paper. Its theorem `not_totallyReal` composes the preceding results.

The minpoly hypothesis in `TwoCycleCertificate` is deliberately intrinsic: it says that the element recovered in the trace field really has minimal polynomial $X^2 - 5$. The exact source certifier supplies this case-specific fact; Lean then derives the bad embedding rather than accepting that embedding as an input.

8.2 Faithful translation of external inputs

The structure `CoxeterExternalInputs` has fields for:

- the case name $P_{11,6}$, $P_{11,7}$, or $P_{11,8}$;
- the assertion that its reflection group is a finite-volume lattice;
- the assertion that K is its adjoint trace field; and
- the Emery–Mila implication from pseudo-arithmeticity to total reality.

The theorem `coxeter_case_main_theorem` proves, from these fields and a `TwoCycleCertificate`, the conjunction

$$\text{finite-volume lattice} \wedge \text{trace-field identification} \wedge \neg \text{total reality} \wedge \neg \text{pseudo-arithmeticity}.$$

Because the imported facts are structure fields supplied locally, the Lean environment gains no axiom from this translation.

Likewise, `ManifoldExternalInputs` names the finite-index relation, torsion-free orientation-preserving subgroup, quotient construction, finite-cover geometry, trace-field invariance, and the correspondence between a quotient and its fundamental group. The theorem

$$\text{finiteIndex_manifold_corollary}$$

checks the complete logical passage to Corollary 1.2.

8.3 Other formal modules

The module `NonTotalReality.lean` contains the core field theorem in a minimal form. The separate module `ConditionalDescent.lean` concerns the supplementary normalized-Clifford and split-support criterion documented in the repository notes. It represents ABHN, local Brauer functoriality, Hasse–Minkowski, and the pseudo-arithmetic candidate classification as ordinary hypotheses and checks their downstream implication graph. That criterion is not needed for Theorem 1.1; the main examples are settled by non-total reality before any Clifford invariant is considered.

8.4 Audit statement

A release audit searches all project Lean sources for the declarations `axiom` and `opaque` and for the proof escapes `sorry`, `admit`, and `native_decide`. None occurs. The trusted computing base is therefore the Lean kernel, the pinned Lean compiler and mathlib declarations, and the explicitly visible hypotheses at the external boundary. The successful command is

```
cd formal && lake build.
```

9 External-input register

This register is normative: an assertion listed here is used but not newly proved by the manuscript or its Lean project. Each item says both what the paper imports and how the Lean theorem exposes it.

E1: Census rows and finite volume.

Ma–Zheng’s classification supplies the three vectors, positive radical values, cusp counts, volumes, and the assertion that they are finite-volume hyperbolic Coxeter 4-polytopes [5]. Exact scripts verify the transcription, Gram rank, and inertia; a direct vertex-link proof independently verifies finite volume for $P_{11,8}$. In Lean this is the field `finite_volume` of `CoxeterExternalInputs`.

E2: Vinberg field identification.

For a finite-volume Coxeter polytope, the field generated by cyclic products in $2G$ is the adjoint trace field of the reflection lattice [8]. In Lean the concrete algebra is a `TwoCycleCertificate`; the geometric identification is the separate field `trace_field`.

E3: Pseudo-arithmetic fields are totally real.

Emery–Mila’s definition and Section 1.4 imply that a pseudo-arithmetic lattice in dimension $n > 3$ has totally real adjoint trace field [2]. Lean translates this as the structure field

```
pseudoArithmetic_totallyReal.
```

It is an implication, so the kernel checks its contrapositive application without proving its geometric premise.

E4: Extension and realization of number-field embeddings.

The standard extension theorem and mathlib’s theorem describing evaluations of number-field embeddings by roots of a minimal polynomial are used to realize $s \mapsto -\sqrt{5}$. The elementary extension proof is included in Proposition 2.3; Lean uses mathlib’s existing verified result rather than a project hypothesis.

E5: Selberg’s lemma.

A finitely generated characteristic-zero linear group has a torsion-free finite-index subgroup [7]. It is not formalized from first principles here. The Lean field `finiteIndex_cover_exists` exposes the precise existence statement needed after also passing to the orientation subgroup.

E6: Finite-index trace-field invariance.

The adjoint trace field is a commensurability invariant [8]. Lean translates the used direction as `traceField_finiteIndex_invariant`.

E7: Finite-cover quotient geometry.

The standard facts that a torsion-free orientation-preserving subgroup gives an orientable manifold, finite index preserves finite volume, and a finite cover of the noncompact quotient remains noncompact are proved at the elementary level in Subsection 2.3. Lean packages their geometric realization as `quotient_geometry`; there is presently no project formalization of hyperbolic orbifolds themselves.

No item E1–E7 is hidden behind a global Lean axiom. Replacing an input by a future library theorem would change the constructor used to build the corresponding structure, not the checked proof downstream.

10 Reproducibility and release audit

The repository is designed so that the central conclusion can be checked at three increasing levels of cost.

Level 1: read the certificate. Check the four entries in (4.5), multiply the cycles, follow Lemmas 5.1 and 5.3, and apply Inputs E2 and E3. No computer algebra is required.

Level 2: build the formal and written proofs. Run

```
cd formal
lake build

cd ../manuscript
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  non_pseudo_arithmetic_4_manifolds.tex
```

The project pins Lean/mathlib, and the TeX source has a fixed date.

Level 3: repeat all exact source checks. Follow REPRODUCIBILITY.md to obtain the pinned Ma-Zheng source and HCPdm incidence file, verify their SHA-256 values, and run each Python or PARI/GP job serially. Reference JSON certificates are committed, so the mathematical outputs remain inspectable without network access. Timing and peak-memory metadata may vary and are not used as mathematical evidence.

The command

```
python3 scripts/release_check.py
```

performs the local release audit: JSON parsing, Python byte-compilation, forbidden-token checks, documentation consistency, Lean build, exact regressions, and TeX build. The packaging script `scripts/build_release.sh` creates a deterministic archive and checksum under `dist/`. The execution record is split between `PRE_RELEASE_CHECKLIST.md` and `RELEASE_NOTES.md`.

A Dependency graph of the main conclusion

For quick auditing, the complete argument is the following chain.

$$\begin{array}{c}
\text{Ma-Zheng row (E1)} \\
\Downarrow \\
\text{four entries of } A = 2G \\
\Downarrow \\
c_{12} = (3 + s)/2, \quad c_{035} = -2u \\
\Downarrow \\
s, u \in k(P), \quad \text{minpoly}_{\mathbb{Q}}(s) = X^2 - 5 \\
\Downarrow \\
(s - 1)u^2 + 2(s - 3)u + s - 7 = 0 \\
\Downarrow \\
\text{the place } s \mapsto -\sqrt{5} \text{ forces } u \notin \mathbb{R} \\
\Downarrow \\
k(P) \text{ is not totally real} \\
\Downarrow \text{ Vinberg (E2)} \\
k(\Gamma) \text{ is not totally real} \\
\Downarrow \text{ Emery-Mila (E3)} \\
\Gamma \text{ is not pseudo-arithmetic.}
\end{array}$$

The manifold corollary appends E5–E7 and E6. The radical relation and both downstream implication chains are checked by Lean; E1–E3 and E5–E7 remain visible external inputs.

B Independent full-field data for comparison

The full-field computation is not part of the proof, but it gives a useful cross-check. The signature $(4, 2)$ of the degree-eight field means that it has four real embeddings and two conjugate pairs of nonreal embeddings, since $r_1 + 2r_2 = 8$. The two-cycle proof detects at least one of those pairs without constructing the other six embeddings. This explains why the short certificate is logically stronger as an audit artifact: it contains exactly the information needed for non-total reality and no dependence on a large primitive-element calculation.

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